CM25D008

PRACTICAL NO :-09

AIM :- - Consider a disk queue with requests for I/O to blocks on cylinders 98,
I83, 41, 122, 14,124,65,67. The SSTF scheduling algorithm is used. The head is
initially at cylinder number 53. The cylinders are numbered from 0 to 199.
Write a Program to calculate the total head movement (in number of cylinders)
incurred while servicing these requests.

```
  GNU nano 8.7
#include <stdio.h>
#include <stdlib.h>

int main() {
    int requests[] = {98,183,41,122,14,124,65,67};
    int n = 8;
    int head = 53;

    int visited[8] = {0};
    int total_movement = 0;

    for(int i = 0; i < n; i++) {
        int min = 9999, index = -1;

        for(int j = 0; j < n; j++) {
            if(!visited[j]) {
                int distance = abs(head - requests[j]);

                if(distance < min) {
                    min = distance;
                    index = j;
                }
            }
        }

        total_movement += min;
        head = requests[index];
        visited[index] = 1;
    }

    printf("Total Head Movement = %d\n", total_movement);

    return 0;
}
```

```
sejal sarode@INBOOK_X2_SLIM MINGW64 ~/OneDrive/Desktop
$ nano sstf.c

sejal sarode@INBOOK_X2_SLIM MINGW64 ~/OneDrive/Desktop
$ gcc sstf.c -o sstf

sejal sarode@INBOOK_X2_SLIM MINGW64 ~/OneDrive/Desktop
$ ./sstf
Total Head Movement = 323
```